INCH-POUND

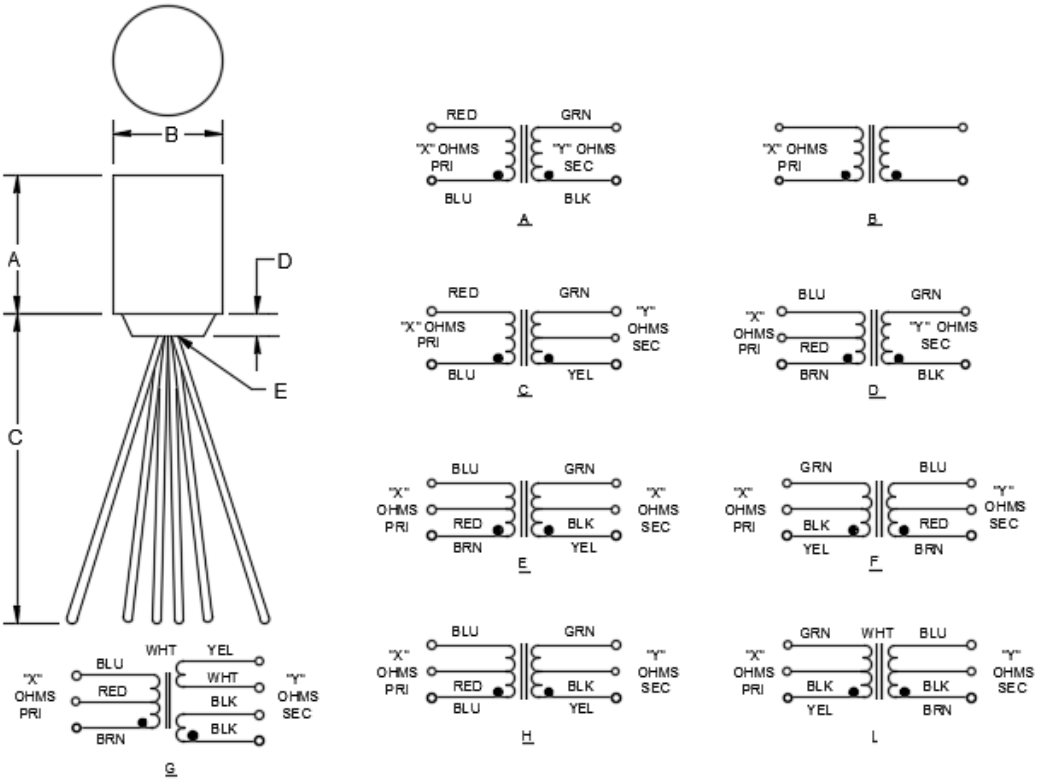
MIL-PRF-27/172E
w/AMENDMENT 2
20 June 2024
SUPERSEDING
MIL-PRF-27/172E
w/AMENDMENT 1
17 March 2017

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY, MICRO-MINIATURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Primary and secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.

AMSC N/A FSC 5950

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Dim	Inches	mm
A	.469 MAX.	11.91 MAX.
B	.344 DIA MAX.	8.74 DIA MAX.
C	1.50 ±.25	38.1 ±6.4
D	.030 MAX.	0.76 MAX.
E	AWG NO.26 (.016)	

FIGURE 1. Dimensions and configurations – continued.

REQUIREMENTS: (When the colors in parentheses, i.e. (Brn-Blu), are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, i.e. (Grn-Yel), the windings are connected in series.)

Electrical ratings: See table I.

Working voltage (peak): 50 V.

Frequency range: 300 Hz to 100 kHz.

Design and construction:

Dimensions and configurations: See figure 1.

Duty cycle: Continuous.

Terminals: Solid wire, insulated. Insulation shall not be polyvinyl chloride.

Composition: D in accordance with MIL-STD-1276.

Diameter: 0.016 inch.

Length: 1.50 inches ±0.25.

Final finish: Tinned.

Weight: 4.50 grams, maximum.

Altitude: 150,000 feet, maximum.

Operating temperature range: -55°C to +105°C

Terminal strength: MIL-STD-202-211, test condition A, 2.0 pounds.

Vibration (high frequency): MIL-STD-202-204, test condition B.

Dielectric withstanding voltage:

At sea level: 100 V rms.

At barometric pressure: 100 V rms.

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Electrical characteristics: See table II.

No load (center tap unbalance): ±1 percent at 1 V, 5 kHz across the primary.

Harmonic distortion: Total harmonic content of the output shall be a maximum of 5 percent at the specified power level and primary dc current (see table I) at 1 kHz to 100 kHz.

Insertion loss: At the specified power level (see table I), the insertion loss shall be a maximum of 3.0 dB at 1 kHz.

Frequency response: ±3 dB at the rated source and load impedances (see table II) with 1.0 mW level output and a reference frequency of 1 kHz.

Self-resonant frequency: Measured with secondaries carrying specified loads, with the secondary voltage observed (see table II)..

Marking location: See figure 1.

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TABLE I. Electrical ratings. 1/ 2/

Dash no.	Circuit diagram	Primary impedance (ohms)		Secondary 3/ impedance (ohms)		Primary dc current (mA) 4/	Power level at 1 kHz (max) (mW)	Primary dc resistance ±25% (ohms)	Secondary dc resistance ±25% (ohms)
01	G	80 CT 100 CT	(Brn-Blu)	32 SPLIT 40 SPLIT	(Yel-Grn)	12 10	600	9.8	6
02	D	120 CT 150 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	10 10	500	10	0.46
03	D	150 CT 200 CT	(Brn-Blu)	12 16	(Blk-Grn)	10 10	500	11	1.5
04	D	300 CT 400 CT	(Brn-Blu)	12 16	(Blk-Grn)	7 7	500	20	1.63
05	D	300 CT	(Brn-Blu)	600	(Blk-Grn)	7	600	19	89
06	D	320 CT 400 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	7 7	500	20	0.46
07	G	400 CT 500 CT	(Brn-Blu)	40 SPLIT 50 SPLIT	(Yel-Grn)	8 6	500	46	8
08	G	400 CT 500 CT	(Brn-Blu)	120 SPLIT 150 SPLIT	(Yel-Grn)	8 6	600	46	26
09	G	400 CT 500 CT	(Brn-Blu)	400 SPLIT 500 SPLIT	(Yel-Grn)	8 6	600	46	74
10	E	400 CT 500 CT	(Brn-Blu)	4,000 CT 5,000 CT	(Yel-Grn)	8 6	600	46	550
11	A	500 600	(Blu-Red)	50 60	(Blk-Grn)	3 3	600	60	8
12	D	500 CT	(Brn-Blu)	600	(Blk-Grn)	5.5	600	31	90
13	A	600	(Blu-Red)	3.2	(Blk-Grn)	3	100	60	0.58
14	D	600 CT 800 CT	(Brn-Blu)	12 16	(Blk-Grn)	5 5	500	43	1.5
15	D	640 CT 800 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	5 5	500	43	0.46
16	D	800 CT 1,000 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	4 4	500	51	0.46
17	D	800 CT 1,070 CT	(Brn-Blu)	12 16	(Blk-Grn)	4 4	500	51	1.5
18	D	900 CT	(Brn-Blu)	600	(Blk-Grn)	4	600	53	89
19	A	1,000 1,200	(Blu-Red)	50 60	(Blk-Grn)	3 3	600	115	8

See footnotes at end of table I.

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TABLE I. Electrical ratings. 1/ 2/ - Continued.

Dash no.	Circuit diagram	Primary impedance (ohms)		Secondary 3/ impedance (ohms)		Primary dc current (mA) 4/	Power level at 1 kHz (max) (mW)	Primary dc resistance ±25% (ohms)	Secondary dc resistance ±25% (ohms)
20	G	1,000 CT 1,200 CT	(Brn-Blu)	16,000 SPLIT 20,000 SPLIT	(Yel-Grn)	3.5 3.5	100	120	940
21	D	1,000 CT 1,330 CT	(Brn-Blu)	12 16	(Blk-Grn)	3.5 3.5	500	71	1.5
22	D	1,060 CT 1,330 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	3.5 3.5	500	71	0.46
23	A	1,200	(Blu-Red)	3.2	(Blk-Grn)	2	600	105	0.58
24	D	1,500 CT 2,000 CT	(Brn-Blu)	12 16	(Blk-Grn)	3 3	500	108	1.5
25	D	1,500 CT	(Brn-Blu)	600	(Blk-Grn)	3	600	86	89
26	D	1,600 CT 2,000 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	3 3	500	109	0.46
27	G	2,000 CT 2,500 CT	(Brn-Blu)	2,000 SPLIT 2,500 SPLIT	(Yel-Grn)	3 3	600	195	125
28	G	2,000 CT 2,500 CT	(Brn-Blu)	8,000 SPLIT 10,000 SPLIT	(Yel-Grn)	3 3	125	195	455
29	D	7,500 CT 10,000 CT	(Brn-Blu)	12 16	(Blk-Grn)	1 1	100	505	1.6
30	D	8,000 CT 10,000 CT	(Brn-Blu)	3.2 4.0	(Blk-Grn)	1 1	100	505	0.46
31	A	10,000	(Blu-Red)	3.2	(Blk-Grn)	1	100	790	0.68
32	C	10,000 12,000	(Blu-Red)	500 CT 600 CT	(Yel-Grn)	1 1	125	780	50
33	C	10,000 12,500	(Blu-Red)	1,200 CT 1,500 CT	(Yel-Grn)	1 1	125 125	780	115
34	E	10,000 CT 12,000 CT	(Brn-Blu)	1,500 CT 1,800 CT	(Yel-Grn)	1 1	125 125	780	126
35	G	10,000 CT 12,000 CT	(Brn-Blu)	2,000 SPLIT 2,400 SPLIT	(Yel-Grn)	1 1	125	560	230
36	C	10,000 12,500	(Blu-Red)	2,000 CT 2,500 CT	(Yel-Grn)	1	125	780	190
37	E	10,000 CT 12,000 CT	(Brn-Blu)	10,000 CT 12,000 CT	(Yel-Grn)	1 1	125 125	975	1,175
38	A	20,000 30,000	(Blu-Red)	800 1,200	(Blk-Grn)	0.5 0.5	60	830	115
39	E	20,000 CT 30,000 CT	(Brn-Blu)	800 CT 1,200 CT	(Yel-Grn)	0.5 0.5	60	830	115
40	G	20,000 CT 30,000 CT	(Brn-Blu)	1,000 SPLIT 1,500 SPLIT	(Yel-Grn)	0.5 0.5	50	800	113
41	G	40,000 CT 50,000 CT	(Brn-Blu)	400 SPLIT 500 SPLIT	(Yel-Grn)	0.25 0.25	50	1,700	60

See footnotes at end of table I.

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TABLE I. Electrical ratings. 1/ 2/ - Continued.

Dash no.	Circuit diagram	Primary impedance (ohms)		Secondary 3/ impedance (ohms)		Primary dc current (mA) 4/	Power level at 1 kHz (max) (mW)	Primary dc resistance ±25% (ohms)	Secondary dc resistance ±25% (ohms)
42	E	40,000 CT 50,000 CT	(Brn-Blu)	4,000 CT 5,000 CT	(Yel-Grn)	0.25 0.25	50	1,700	450
43	B	200,00	(Blk-Grn)	1,000	(Blu-Red)	0	25	9,000	100
44	F	200,000 CT	(Yel-Grn)	1,000 CT	(Brn-Blu)	0	25	9,000	100
45	E	600 CT	(Brn-Blu)	600 CT	(Yel-Grn)	4	600	47	47
46 5/	H	4,000 CT 5,000 CT	(Brn-Blu)	8,000 CT 10,000 CT	(Yel-Grn)	2 2	100	320	590
47 5/	H	8,000 CT 10,000 CT	(Brn-Blu)	1,200 CT 1,500 CT	(Yel-Grn)	1 1	100	640	110
48 5/	H	9,000 CT 10,000 CT	(Brn-Blu)	9,000 CT 10,000 CT	(Yel-Grn)	1 1	100	850	1,080
49 5/	I	100,000 CT	(Yel-Grn)	500 CT	(Brn-Blu)	0	25	7,900	85
50 5/	I	200,000 CT	(Yel-Grn)	1,000 CT	(Brn-Blu)	0	25	10,700	100

1/ Qualification testing and approval to M27/172-49 or M27/172-44 shall be sufficient to grant qualification approval to M27/172-01 through M27/172-50.
2/ Impedance values written one above the other indicates a range of matching impedances over which the parts will give satisfactory performance as long as the impedance ratio is maintained.
3/ Where windings are listed as SPLIT, one-fourth of the listed impedance is available by paralleling the winding.
4/ Primary dc current shall be the maximum single-ended current for which the transformer will meet the specified performance requirements. For push-pull, mA dc can be any balanced value taken by .5 W transistors.
5/ Includes electrostatic shield. Shield terminal shall be white. Voltage ratio shall be 2 to 1 at 20 kHz.

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TABLE II. Electrical characteristics.

Dash no.	Frequency response: ±3 dB 1/ at 300 Hz to 100 kHz and 1 mW		Resonance, self resonant frequency (min) (kHz)	Polarity: Additive with terminals (below) connected
	Z_S (ohms)	Z_L (ohms)		
01	80 CT (Brn-Blu)	32 (Yel-Grn)	1,500	Blu and Yel, and Wht and Blk
02	120 CT (Brn-Blu)	3.2 (Blk-Grn)	1,000	Blu and Blk
03	150 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
04	300 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
05	300 CT (Brn-Blu)	600 (Blk-Grn)	750	Blu and Blk
06	320 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk
07	400 CT (Brn-Blu)	40 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
08	400 CT (Brn-Blu)	120 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
09	400 CT (Brn-Blu)	400 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
10	400 CT (Brn-Blu)	4,000 CT (Yel-Grn)	750	Blu and Yel
11	500 (Blu-Red)	50 (Blk-Grn)	750	Red and Blk
12	500 CT (Brn-Blu)	600 (Blk-Grn)	750	Blu and Blk
13	600 (Blu-Red)	3.2 (Blk-Grn)	750	Red and Blk
14	600 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
15	640 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk
16	800 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk
17	800 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
18	900 CT (Brn-Blu)	600 (Blk-Grn)	750	Blu and Blk
19	1,000 (Blu-Red)	50 (Blk-Grn)	750	Red and Blk
20	1,000 CT (Brn-Blu)	16,000 (Blk-Grn)	750	Blu and Yel, and Wht and Blk
21	1,000 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
22	1,060 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk

See footnote at end of table II.

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TABLE II. Electrical characteristics – Continued.

Dash no.	Frequency response: ±3 dB 1/ at 300 Hz to 100 kHz and 1 mW		Resonance, self resonant frequency (min) (kHz)	Polarity: Additive with terminals (below) connected
	Z_S (ohms)	Z_L (ohms)		
23	1,200 (Blu-Red)	3.2 (Blk-Grn)	750	Red and Blk
24	1,500 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
25	150 CT (Brn-Blu)	600 (Blk-Grn)	750	Blu and Blk
26	1,600 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk
27	2,000 CT (Brn-Blu)	2,000 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
28	2,000 CT (Brn-Blu)	8,000 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
29	7,500 CT (Brn-Blu)	12 (Blk-Grn)	750	Blu and Blk
30	8,000 CT (Brn-Blu)	3.2 (Blk-Grn)	750	Blu and Blk
31	10,000 (Blu-Red)	3.2 (Blk-Grn)	750	Red and Blk
32	10,000 (Blu-Red)	500 CT (Yel-Grn)	750	Red and Yel
33	10,000 (Blu-Red)	1,200 CT (Yel-Grn)	750	Red and Yel
34	10,000 CT (Brn-Blu)	1,500 CT (Yel-Grn)	750	Blu and Yel
35	10,000 CT (Brn-Blu)	2,000 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
36	10,000 (Blu-Red)	2,000 CT (Yel-Grn)	750	Red and Yel
37	10,000 CT (Brn-Blu)	10,000 CT (Yel-Grn)	500	Blu and Yel
38	20,000 (Blu-Red)	800 (Blk-Grn)	750	Red and Blk
39	20,000 CT (Brn-Blu)	800 CT (Yel-Grn)	750	Blu and Yel
40	20,000 CT (Brn-Blu)	1,000 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
41	40,000 CT (Brn-Blu)	400 (Yel-Grn)	500	Blu and Yel, and Wht and Blk
42	40,000 CT (Brn-Blu)	4,000 (Yel-Grn)	500	Blu and Yel
43	200,000 (Blk-Grn)	1,000 (Blu-Red)	100	Grn and Blu
44	200,000 CT (Yel-Grn)	1,000 CT (Brn-Blu)	100	Blu and Yel

See footnote at end of table II.

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TABLE II. Electrical characteristics – Continued.

Dash no.	Frequency response: ±3 dB 1/ at 300 Hz to 100 kHz and 1 mW		Resonance, self resonant frequency (min) (kHz)	Polarity: Additive with terminals (below) connected
	Z_S (ohms)	Z_L (ohms)		
45	600 CT (Brn-Blu)	500 CT (Blk-Grn)	100	Blu and Yel
46	4,000 CT (Brn-Blu)	8,000 CT (Blk-Grn)	500	Blu and Yel
47	8,000 CT (Brn-Blu)	1,200 CT (Blk-Grn)	500	Blu and Yel
48	9,000 CT (Brn-Blu)	9,000 CT (Blk-Grn)	500	Blu and Yel
49	100,000 CT (Yel-Grn)	500 CT (Yel-Grn)	100	Blu and Yel
50	200,000 CT (Yel-Grn)	1,000 CT (Yel-Grn)	100	Grn and Brn

1/ Frequency response shall be within ±3 dB at 300 Hz to 20 kHz and ±4 dB at 300 Hz to 100 kHz at 1 mW for dash numbers 43, 44, 49, and 50.

Part or Identifying Number (PIN): M27/172-(dash number from table I).

SUPERSESSION DATA:

MIL-PRF-27/172 supersedes the following MS sheets which have been cancelled:

MS21375	MS21407	MS21411
MS21377	MS21408	MS21412
MS21393	MS21409	MS21413
MS21406	MS21410	MS21421
		MS53228

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MIL-PRF-27/172 supersedes MIL-PRF-27/28 and MIL-PRF-27/144. Supersession data are as follows:

MIL-PRF-27/172 PIN	Superseded PIN
M27/172-46	M27/28-01
M27/172-47	M27/28-02
M27/172-48	M27/28-03
M27/172-49	M27/28-04
M27/172-50	M27/28-05
M27/172-01	M27/144-01
M27/172-05	M27/144-02
M27/172-07	M27/144-03
M27/172-09	M27/144-04
M27/172-10	M27/144-05
M27/172-11	M27/144-06
M27/172-12	M27/144-07
M27/172-45	M27/144-08
M27/172-18	M27/144-09
M27/172-19	M27/144-10
M27/172-23	M27/144-11
M27/172-25	M27/144-12
M27/172-27	M27/144-13
M27/172-28	M27/144-14
M27/172-32	M27/144-15
M27/172-33	M27/144-16
M27/172-34	M27/144-17
M27/172-35	M27/144-18
M27/172-36	M27/144-19
M27/172-37	M27/144-20
M27/172-38	M27/144-21
M27/172-39	M27/144-22

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204, MIL-STD-202-211, MIL-STD-1276

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, CR4
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5950-2024-019)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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